

Mohammad Zafari

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EDUCATION

Kharazmi University

Tehran, Iran

B.Sc. in Computer Science; GPA: 3.86/4.0 (WES)

Sep 2019 – July 2024

Ranked 3rd in class of 40

Relevant Coursework: Linear Algebra (A⁺), Introduction to Probability and Statistics (A⁺), Probability Theory (A⁺), Graph Theory (A⁺), Calculus III (A⁺), Numerical Analysis (A), Abstract Algebra (B), Combinatorics (A⁺), Logic and Set Theory (A⁺), Discrete Mathematics (B), Real Analysis (A⁺), Artificial Intelligence (A⁺), Design and Analysis of Algorithms (A⁺), Topics in Computer Science (Machine Learning) (A⁺), Senior Capstone Project (A⁺), Object-Oriented Programming (A), Nonlinear Optimization (A⁺)

RESEARCH AND TEACHING EXPERIENCE

Kharazmi University

Tehran, Iran

Research Assistant

Feb 2023 – July 2023

- Developed knowledge distillation framework for image classification using PyTorch, inspired by recent advances in attention-based feature matching, improving student performance on the Stanford40 and CIFAR100 datasets. This project uses an attention mechanism to identify effective feature pairs for distillation. This project was under the supervision of [Prof. Mohammad Soltanian](#).[↗](#)

Kharazmi University

Tehran, Iran

Teaching Assistant

Feb 2022 – May 2022

- Conducted recitation sessions and actively monitored student progress to identify challenging topics in [Theory of Computation](#) under the supervision of [Dr. Arash Ahadi](#).[↗](#)

Kharazmi University

Tehran, Iran

Teaching Assistant

Sep 2021 – Dec 2021

- Graded assignments and exams, provided academic support by answering student questions and explaining course material in [Introduction to Probability and Statistics](#) under the supervision of [Prof. Meisam Moghimbeygi](#).[↗](#)

SELECTED PROJECTS

Knowledge Distillation For Image Classification | [GitHub](#) [↗](#)

- As part of the [Senior Capstone Project](#), I developed various knowledge distillation networks. Specifically, I used attention-based feature distillation techniques for two image classification datasets: Stanford40 and CIFAR100. I utilized [custom data loaders](#) [↗](#) for efficient data handling. I used different teacher and student models (e.g., ResNet-34 to ResNet-18, and WideResNet-40-2 to WideResNet-16-2) and trained them using the distillation technique. My focus was on improving the student model's accuracy while keeping it lightweight. This project is implemented in **PyTorch**.

Clustering Algorithms Comparison | [GitHub](#) [↗](#)

- As part of the [Topics in Computer Science \(Machine Learning\)](#) project, I compared three clustering algorithms: K-Means, Agglomerative Clustering, and DBSCAN. These models were applied to three synthetic datasets: noisy circles, noisy moons, and isotropic Gaussian blobs, which evaluate their performance on different types of data. I visualized the clustering results and assessed model performance using several evaluation metrics, including the Rand Index, Jaccard Index, Silhouette Score, and Dunn Index. This project enhanced my understanding of unsupervised learning techniques and provided practical experience in clustering and model evaluation. This project is implemented using the **Scikit-Learn** library.

Digit Recognition With MNIST Dataset | [GitHub](#) [↗](#)

- As part of [Topics in Computer Science \(Machine Learning\)](#) project, I implemented multilayer perceptron (MLP) models with three fully connected layers, using a variety of activation functions including ReLU, SELU, ELU, and Tanh. The networks were trained with multiple optimizers such as SGD, Adam, RMSprop, AdamW, Adadelata, Adagrad, Adamax, Adafactor, Nadam, and FTRL. Implementations were done in both **Scikit-Learn** and **TensorFlow** to compare the behavior and performance between the two libraries.

Compare Optimization Algorithms | [GitHub](#)

- As part of the [Nonlinear Optimization](#) project, I implemented and compared optimization algorithms—specifically Gradient Descent and Newton’s Method—on multivariable nonlinear functions using **Python**. This project involved computing gradients and Hessians, analyzing convergence behavior, and visualizing optimization paths to explore how different initial conditions and step sizes affect results.

Implementing a Hotel Chain Database | [GitHub](#)

- As part of the [Database Systems](#) project, I Designed and implemented a relational database system for hotel chain management using **SQL**. The schema included 15 interrelated tables covering branches, rooms, customers, bookings, and staff. I wrote SQL queries for data retrieval, applied normalization to reduce redundancy, and enforced data integrity using primary and foreign keys. Each room also has different types, based on which they can have different prices. Each employee can have different roles. Also, each employee has a supervisor and each branch has a manager. Each customer can reserve a number of rooms.

Candy Store Simulation | [GitHub](#)

- As part of the [Computer Simulation and Modeling](#) project, I used the SimPy discrete-event simulation library in **Python** to model the operation of a candy store. The simulation included customers arriving at random normal distribution intervals, waiting in queues, being served by multiple cashiers, and tracking key performance metrics such as average wait times and queue lengths. Furthermore, a second candy store scenario included high-priority customers who were served immediately upon arrival.

15-Puzzle Solver | [GitHub](#)

- As part of the [Computer Simulation and Modeling](#) project, which solves and visualizes the 15-puzzle problem using search algorithms such as DFS, Hill-Climbing and A*, to explore various solution strategies. I have implemented Hill-Climbing and A*, while the visualization component was developed by my teammate. This project is Implemented by **Python**.

Hamilton Night’s Tour Solver | [GitHub](#)

- As part of the project for [Object-Oriented Programming](#) course, I developed a **Java**-based solution to the classic Knight’s Tour problem, where the objective is to move a knight on a chessboard such that it visits every square exactly once. The implementation utilizes Warnsdorff’s Rule—which selects the next move based on the square with the fewest onward moves—to efficiently find a valid tour. The algorithm dynamically calculates the accessibility of each square and selects the optimal path to minimize backtracking.

SPSS Project | [Report](#)

- As part of the [Introduction to Probability and Statistics](#) project, I analyzed statistical parameters in an SPSS dataset, which shows the amount of aflatoxin in 8 types of food. The analysis involved calculating descriptive statistics—such as mean, mode, median, and standard deviation—for each type of food. Additionally, I generated frequency distribution tables and assessed indices of central tendency and dispersion to understand the variability and distribution patterns of aflatoxin levels within each food category.

STANDARD EXAMS

TOEFL iBT Overall: **91/120**

Attempted 2025

Reading: 21/30

Listening: 23/30

Speaking: 22/30

Writing: 25/30

SKILLS

Languages: C, Java, Python, SQL, MATLAB, Assembly

Frameworks: PyTorch, TensorFlow, Scikit-learn, OpenCV

CERTIFICATES

Supervised Machine Learning: Regression and Classification | [Coursera](#) 

Advanced Python Programming and Object-Oriented Thinking Course | [Quera](#) 

Task-Oriented Course in Fundamentals of Python Programming and Algorithmic Thinking | [Quera](#) 

RELEVANT COURSEWORK

Course	Grade	Course	Grade
Calculus I	A ⁺	Programming Fundamentals	A
Calculus II	B	Object-Oriented Programming	A
Calculus III	A ⁺	Data Structures and Algorithms	B
Introduction to Probability and Statistics	A ⁺	Design and Analysis of Algorithms	A ⁺
Probability Theory	A ⁺	Nonlinear Optimization	A ⁺
Graph Theory	A ⁺	Artificial Intelligence	A ⁺
Abstract Algebra	B	Computer Simulation and Modeling	A ⁺
Discrete Mathematics	B	Senior Capstone Project	A ⁺
Logic and Set Theory	A ⁺	Theory of Computation	A ⁺
Combinatorics	A ⁺	Operating Systems	A ⁺
Linear Algebra	A ⁺	Computer Architecture	A ⁺
Numerical Linear Algebra	A	Database Systems	A ⁺
Numerical Analysis	A	Topics in Computer Science (Machine Learning)	A ⁺
Real Analysis	A ⁺	Computer Networks	A ⁺
Ordinary Differential Equations	B ⁺	Scientific Computing	A ⁺
Digital Logic Design	A	Compiler Design	A ⁺

REFERENCES

Mohammad Soltanian, *Associate Professor* [!\[\]\(4729e517bc6a7cd81c8025b9646574fb_img.jpg\)](#)

Department of Computer Science,
Kharazmi University

Meisam Moghimbeygi, *Assistant Professor* [!\[\]\(a03a7eb2f4046e1d3c76772003e549ea_img.jpg\)](#)

Department of Mathematics,
Kharazmi University

Arash Ahadi, *Associate Professor* [!\[\]\(e474458956c9a37fbf9586ddb60a7fa1_img.jpg\)](#)

Department of Computer Science & Mathematics,
Kharazmi University

Keivan Borna, *Assistant Professor* [!\[\]\(5361750c22c4e047a52f4eac1ec2d4cc_img.jpg\)](#)

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